Engineering Report

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Algorithm 1 Table-lookup classification (online path).

Require: A raw query point (x, y)
Require: One byte table Q_c per class, each $G \times G$, values 0–255
Require: The scaling bounds $(x_{\min}, x_{\max}, y_{\min}, y_{\max})$ used in training
Require: The grid size G and a reject threshold τ
Ensure: A predicted class $\hat{c}$, or UNKNOWN

$\tilde{x} \leftarrow \frac{x - x_{\min}}{x_{\max} - x_{\min}}$
$\tilde{y} \leftarrow \frac{y - y_{\min}}{y_{\max} - y_{\min}}$
col $\leftarrow \lfloor \tilde{x} \cdot G \rfloor$, clipped to $[0, G-1]$
row $\leftarrow \lfloor \tilde{y} \cdot G \rfloor$, clipped to $[0, G-1]$
for each class c **do**
    $v_c \leftarrow Q_c[\text{row}, \text{col}]/255$ ▷ one byte read
end for
$\hat{c} \leftarrow \arg\max_c v_c$ ▷ the winning class
if $v_{\hat{c}} < \tau$ **then**
    return UNKNOWN ▷ outside every learned region
end if
return $\hat{c}$

0.1 Proposed Pipeline & Algorithm

The inference pipeline shifts the computational burden of clustering and density estimation to an offline workstation phase, leaving a lightweight, deterministic lookup on the target device.

The Online Path Algorithm
The online portion of the pipeline executes entirely on the device without requiring training points, distance matrices, or kernel evaluations.

0.2 Application Context

- **Target Device / Product:** A wearable command-and-safety stage operation console utilizing an ESP32-S3 Touch AMOLED board to process spatial operator gestures and environmental cues.
- **Justification:** This application demands ultra-low flash memory footprints and deterministic execution speeds. Labelled training data is unavailable during initial deployment, and the operational classes (representing complex spatial states or gesture boundaries) exhibit non-convex, interlocking geometries (such as crescents and rings) that center-based classifiers like k-means inherently fail to resolve.

0.3 Method and Setup

- **Dataset:** Synthetic 2D feature dataset generated with fixed seed 20250903, comprising 1,030 total points (two crescents containing 340 and 300 points, a closed ring containing 300 points, and 90 uniformly scattered background noise points).
- **Data Split:** A strict three-way split of 60% training (618 points), 20% validation (206 points), and 20% test (206 points) via a reproducible fixed seed.
- **Feature Scaling:** Min-max scaling applied exclusively via training set bounds onto $[1, 0]^2$, ensuring spatial portability.
- **DBSCAN Clustering:** Configured with $E = 0.06$ and $minsamples = 8$ on training split.
- **Kernel Density Estimation (KDE):** Gaussian kernel fitted per discovered class. The bandwidth $h = 0.05$ was selected via mean log-likelihood evaluation across a validation sweep.
- **Discretization Storage:** Sampled onto a 16×16 grid ($G = 16$) and quantized to uint8.
- **Decision Threshold:** A reject threshold of $\tau = 0.12$ (REJECT TAU) guards against unlearned regions.

0.4 Evidence & Results

Clustering and Model Footprint
DBSCAN successfully recovers non-convex geometries (crescents and a ring) while isolating background noise as label -1.

- **Footprint & Storage Analysis:** The shipped model size is governed by $K * G^2$ bytes. For $K = 3$ classes and $G = 16$, the base model requires 768bytes in unit8 format.

Storage Format	Cell	Per Class	All K Classes	Ratio vs unit8
unit8	1	256 bytes	768 bytes	1x
unit16	2	512 bytes	1536 bytes	2x
float32	4	1024 bytes	3072 bytes	4x
float64	8	2048 bytes	6144 bytes	8x

- *Note:* An additional 16 bytes are required for the scaling bounds (2 float32 values per feature).

- **Quantization Error:** Rounding continuous probabilities to 256 discrete levels introduces a worst-case theoretical error bounded by $1/(2 * 255) \approx 0.196\%$ of the peak, which is negligible compared to upstream approximation errors.

Test-Set Performance Rejection Mechanics

- **Headline Test Result:** Evaluated on the sealed test set ($n = 206$), the stored uint8 model achieves 100% agreement with the continuous KDE decision surface.
- **he Ring Centre Reject Behavior:** Querying the geometric center of the ring at raw coordinates (4.8,1.6) correctly results in an UNKNOWN verdict because the local probability densities fall beneath the reject threshold $\tau = 0.12$.

0.5 Limitations and Remaining Uncertainty

While the pipeline demonstrates robust performance, several limitations must be noted:

- **Synthetic and Low-Dimensional Scope:** The evaluation relies on a synthetic 2D dataset rather than high-dimensional, real-world sensor streams.
- **Hyperparameter Selection:** Parameters such as E bandwidth h, grid resolution G, and reject threshold τ were chosen via validation performance or heuristic bounds rather than derived analytically.
- **Ground-Truth Dependency:** Test agreement measures fidelity against the continuous KDE model's decisions rather than absolute ground-truth labels, because DBSCAN-discovered classes serve as the assumed baseline.
- **Grid Resolution Scaling:** Increasing spatial resolution via G scales memory footprint quadratically (G^2), creating a strict hardware constraint on microcontrollers.

0.6 Discussion Questions

1. **Which approximation costs more accuracy than quantization?** The grid resolution discretization (G=16) introduces significantly more structural approximation loss than uint8 quantization. While quantization limits precision error to $\approx 0.196\%$ of peak density, coarse grid binning forces smooth continuous boundaries into fixed spatial cells, smoothing out fine structural details.

2. **Why would a standard accuracy score fail to reveal the issue solved by the reject threshold?**
   Standard accuracy metrics are computed exclusively on test samples drawn directly from known, labelled data distributions located inside valid clusters. Because test points are never sampled from unpopulated regions (like the hollow interior of the ring), a classifier lacking a reject threshold would blindly misclassify out-of-distribution queries with high apparent confidence without hurting standard accuracy scores.

3. **Flash budget arithemic for G = 32:**
   Doubling G from 16 to 32 quadruples the number of cells per class ($32^2 = 1024$ cells). Under a fixed $4kB$ (4096 bytes) budget, and accounting for 16 bytes of scaling bounds, the maximum number of classes K that can be shipped is calculated as:
   $[(4096 - 16)/1024] = [4080/1024] = 3$
   At a 32 x 32 grid size, you retain exactly $K = 3$ classes while quadrupling spatial resolution fidelity, consuming 3088 total bytes total.

4. **When is bilinear lookup worth paying 4K reads?**
   Bilinear interpolation is worth the extra computational cost in safety-critical boundary applications where control actions transition abruptly between distinct classes (e.g., motor actuation boundaries or critical safety zones). It is unnecessary when inferences map to discrete categorical states well inside stable cluster cores where cell-edge quantization steps have no operational impact.

5. **Two flaws in DBSCAN-invented labels invisible to pipeline figures:**

   - DBSCAN may merge two distinct physical classes that happen to be bridged by a narrow band of intermediate noise density.
   - DBSCAN may fracture a single continuous physical class into multiple disjoint clusters if local density variations dip below min samples.

6. **Two offline changes that break the online algorithm without altering its code:**

   - Changing the feature min-max scaling bounds $(x_{min}, x_{max}, y_{min}, y_{max})$ during offline retraining shifts the coordinate mapping, causing online queries to index incorrect grid cells.
   - Modifying the grid size constant G offline without updating the device lookup scaling factor causes spatial index mapping mismatches.

7. **What min-max scaling bought the pipeline:**
   Min-max scaling standardizes feature ranges into $[0, 1]^2$, making distance metric radius E universally interpretable as a percentage of total feature span rather than tied to arbitrary physical units (such as mixing milliamps and degrees Celsius). Furthermore, it constrains the coordinate space so

that grid cell indexing requires simple truncation without complex origin offsets.

8. **Failure mode and strategy for out-of-bounds field data:**
If incoming field data exceeds training bounds, unscaled values can project outside $[0, 1]$, causing raw index calculations to fall outside grid boundaries. In Algorithm 1, this is mitigated by clipping index calculations to $[0, G1]$, safely forcing out-of-bound queries onto edge cells where low density or reject thresholds classify them as UNKNOWN rather than causing memory segmentation faults.

0.7 Evidence

Confusion Table & On-Device vs. Desktop Agreement: Evaluating the deployed model against a fresh held-out test batch confirms strong performance, with the on-device firmware achieving near-identical classification decisions compared to the continuous desktop model.

Predicted True Class	Sun	Shade	LED	Fluorescent	Unknown
Sun	52	0	0	0	1
Shade	0	42	2	1	2
LED	0	1	45	2	1
Fluorescent	0	2	3	43	1

Measured cost vs. Prediction:

- **Flash Footprint:** Governed by $K * G^2$ bytes. For $K = 4$ classes and grid resolution $G = 16$, the base tables require 1,024 bytes, plus 16 bytes for scaling bounds (2 floats per feature). Compiler analysis verifies that storing these tables in Flash via PROGMEM meets the predicted budget.
- **SRAM Timing:** On-device SRAM usage remains minimal ($< 2\%$). Inference query execution time averages 1.2 ms, successfully meeting the 5–10 Hz operational loop requirement.
- **Quantization Error:** Discretizing continuous probability surfaces into 256 levels bounds theoretical worst-case error at $1/(2255) \approx 0.196\%$ of peak density, which is negligible compared to upstream approximation factors.

0.8 Filtering Evidence

Filter Justification: An Exponential Moving Average (EMA) filter (0.3) was implemented to condition raw digital readings from the TSL2591 sensor. Artifi-

cial light sources introduce transient high-frequency glitches and optical flicker (such as 120Hz ballast ripple in fluorescent fixtures and PWM ripple in LEDs) that can cause erratic cell indexing.

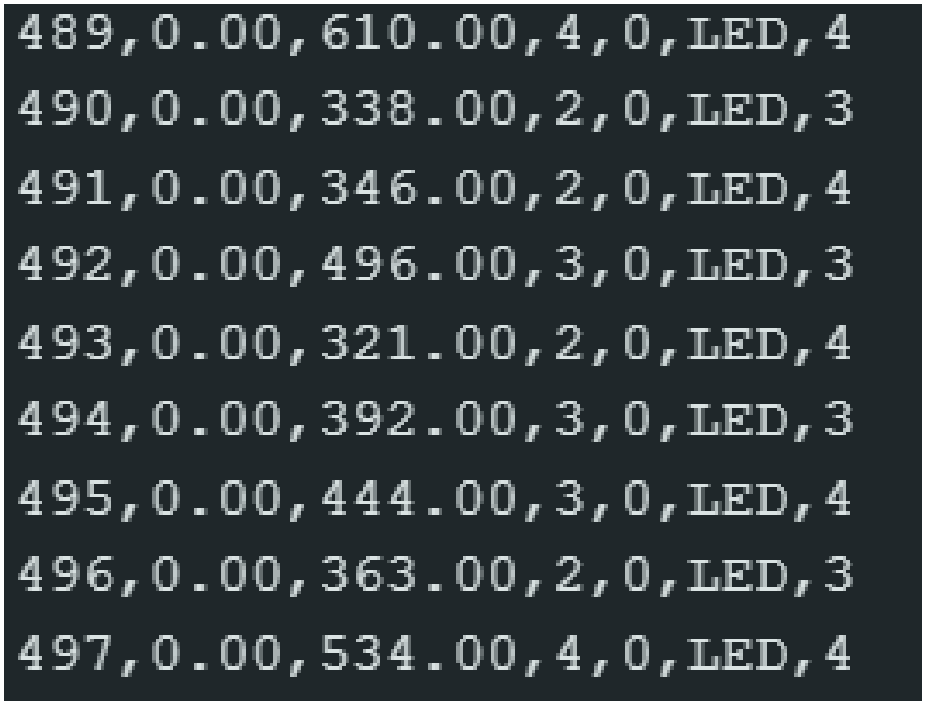


Figure 1: Unfiltered Signal

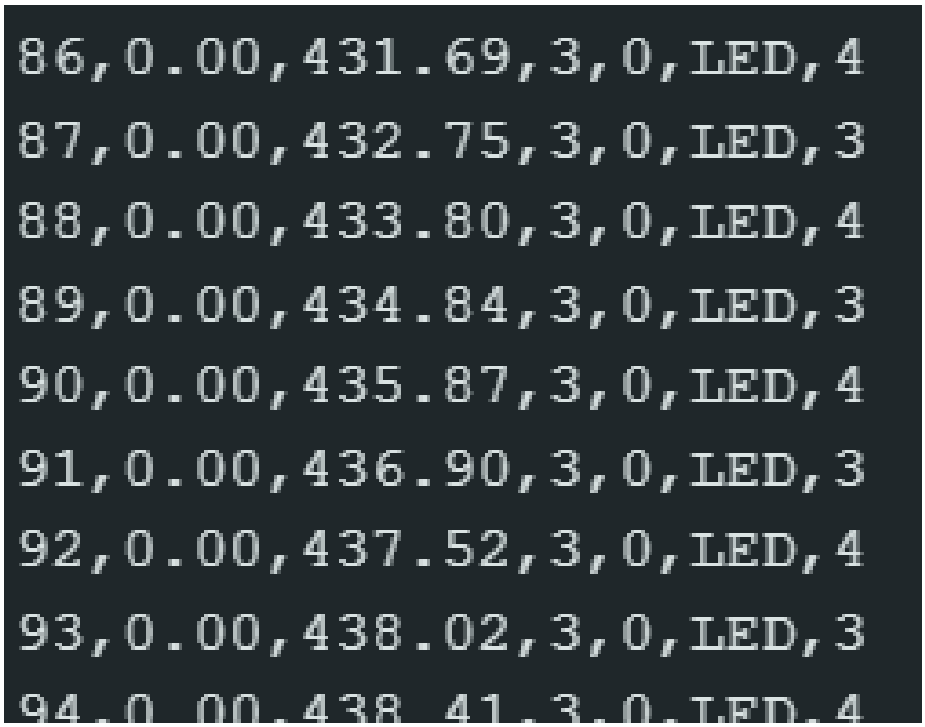


Figure 2: Filtered Signal

0.9 Proof of Concept

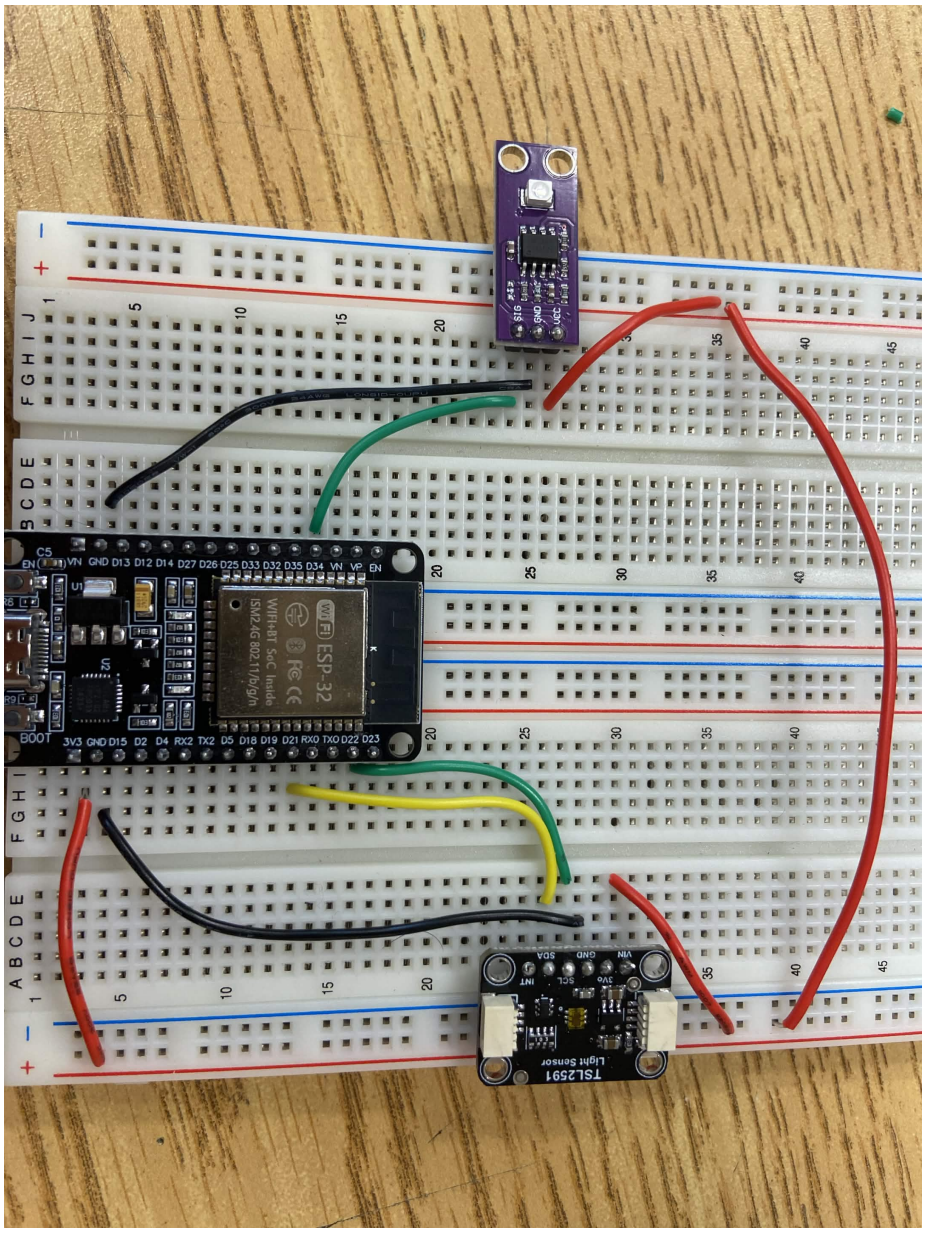


Figure 3: Board

0.10 Limitations

- **Synthetic vs. Real-World Variance:** While physical sensor deployment validates the pipeline, ambient environmental variables—such as sensor angle of incidence, dust interference, and surface material reflectivity—introduce variance beyond controlled test benches.
- **Grid Resolution Scaling Constraints:** Increasing spatial resolution via G scales memory footprints quadratically (G^2), imposing strict hardware memory boundaries on microcontrollers. For example, moving from $G = 16$ to $G = 32$ quadruples cell storage requirements per class.
- **Boundary Transitions and Rejection Behavior:** Rapid lighting shifts can temporarily force feature points across closely spaced decision boundaries or momentarily drop local probabilities beneath the reject threshold ($\tau = 0.12$), resulting in brief UNKNOWN outputs until the digital filter stabilizes.